

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-ad03c75e6032d952c.jsonl`  
- SHA-256: `fe3380fd062a3d997241f99420c3491dac4d2346db5fc495e8e69f5e29fe23ee`  
- Source modified: `2026-07-04T00:53:28+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `subagents`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-04T00:51:13.147Z)

Code-review finder. Target: PR #62 of sports-data-collector (C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector). Diff: `cd <repo> && gh pr diff 62`. PR branch = origin/codex/a3-golden-source-videos. It expands data/golden_source_videos.json to 15 videos to reconcile with the badminton-highlight-indexer eval corpus.

ANGLE ? cross-repo consistency. This manifest is CONSUMED by badminton-highlight-indexer PR #458's `backend/eval/fusion\_golden.py::load\_source\_video\_paths`/'`attach\_source\_video\_paths`, which join on `video\_id` == the eval spec `name` and filter on `source\_status == "present\_local"` + a non-empty `local\_path`. The eval corpus names live in `C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/output/human\_lovo\_manifest.json` (a list of {name, trajectory, labels, ...}).
Check:

1. ****video_id ? eval name join****: extract the 15 `video\_id`s from the PR-branch golden_source_videos.json and the 15 `name`s from human_lovo_manifest.json (`python -c` to load both). Do they match EXACTLY as sets? Any video_id that won't join (typo, case mismatch ? e.g. "GX010128" vs "gx010128", "Boxhill_Doubles" vs "boxhill_doubles")? A mismatch means #458's attach silently skips that video (0 paths attached for it).
2. ****The `annotation\_proxy` MD5 vs the eval's actual file****: #458's consumer uses `local\_path` to run person/audio detection. If the recorded proxy file differs in resolution/fps from the file the golden trajectory/labels were made on, the frame-indexed features won't align. Is there evidence the eval trajectories were made from the SAME file recorded here? (Check if human_lovo_manifest trajectory/labels filenames hint at proxy vs original.)
3. ****A7-doc contradiction****: badminton-highlight-indexer's docs/ALPHA_LAUNCH_READINESS.md (in PR #458) A7 record claims "The source manifest supplied all 15 source-video paths, including the F-drive Boxhill originals." But this manifest marks all 15 present_local. The #458 PR BODY earlier and this #62 PR body say GX010137/GX010141 were missing. Is the "all 15 supplied" claim consistent with this manifest, or does one doc overstate?

Only flag concrete cross-repo breaks. Return up to 6 candidates as JSON [{file, line, summary, failure_scenario}]. [] if none. Final message = only the JSON array.

2. assistant (2026-07-04T00:51:14.932Z)

I'll analyze this cross-repo consistency issue systematically. Let me start by examining both the PR diff and the manifest files.

3. user (2026-07-04T00:51:16.625Z)

```
diff --git a/README.md b/README.md  
index 4956ad2..98b7882 100644  
--- a/README.md  
+++ b/README.md  
@@ -104,7 +104,7 @@ python -m src.cli.curate export --sport badminton --include-staging  
```
```

The exporter writes `//.csv` (decimal-second `start,end` rows) and a `manifest.json` pairing each CSV with its video `local\_path`/`video\_url`, fps, resolution, and

license. The indexer keys videos by **file MD5**, not by our `video_id`, so the downstream eval/training run must process the exact video file named in the manifest. Flags: `--sport`` (restrict to one sport), `--include-staging`` (default: curated only), `--include-noncommercial`` (default: commercial-safe only).

-The six owned golden source MP4s have a committed integrity record at `[`data/golden_source_videos.json`](data/golden_source_videos.json)`. It records their current ``F:`` source paths, byte sizes, and full-file MD5s; uploading those large originals remains owner-gated.

+The 15-video badminton golden eval corpus has a committed source/proxy integrity record at `[`data/golden_source_videos.json`](data/golden_source_videos.json)`. It records current local source paths, byte sizes, and full-file MD5s; uploading those large originals/proxies remains owner-gated.

---

```
diff --git a/data/golden_source_videos.json b/data/golden_source_videos.json
index 6d26348..9db568e 100644
```

```
--- a/data/golden_source_videos.json
```

```
+++ b/data/golden_source_videos.json
```

```
@@ -1,57 +1,181 @@
```

```
{
- "schema": "sports-data-collector.golden_source_videos.v1",
- "generated_at_utc": "2026-07-02T15:08:00Z",
- "purpose": "Durable MD5/path record for the six owned golden source MP4s. The indexer joins
videos by full-file MD5; uploads remain owner-gated.",
- "source_root": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded Videos\\Training\\Golden
Labelled",
+ "schema": "sports-data-collector.golden_source_videos.v2",
+ "generated_at_utc": "2026-07-03T19:53:33Z",
+ "purpose": "Durable source-video/proxy MD5 record for the 15-video badminton golden eval
corpus. The indexer joins videos by full-file MD5; uploads remain owner-gated.",
+ "eval_manifest": "badminton-highlight-indexer/output/human_lovo_manifest.json",
+ "upload_status": "not_uploaded_owner_gated",
+ "source_roots": {
+ "original_six": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled",
+ "annotation_drive": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request
- June 15",
+ "boxhill_runs": "G:\\My Drive\\Sharing\\Badminton\\Boxhill\\2026-06-17 Local no-AI
(improved windowing)"
+ },
+ "videos": [
+ {
+ "video_id": "adarsh_avi_singles",
+ "filename": "Adarsh and Avi.MP4",
+ "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled\\Adarsh and Avi.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
+ "bytes": 11445052772,
+ "last_write_time": "2026-06-01T08:38:28.0000000+05:30",
+ "md5": "751b6266be89dce2e9a39946caa9325c"
+ },
+ {
+ "video_id": "kushagra_singles",
+ "filename": "KushagraYashAviFirstVideo.MP4",
+ "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled\\KushagraYashAviFirstVideo.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
+ "bytes": 8058214679,
+ "last_write_time": "2026-05-22T08:02:16.0000000+05:30",
```

```

- "md5": "1f8ab4ba3eea45fe43a14366f825fc2f"
+ "md5": "1f8ab4ba3eea45fe43a14366f825fc2f",
+ "md5_source": "Get-FileHash"
 },
 {
 "video_id": "largetest_doubles",
 "filename": "LargeTestVideo.MP4",
 "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled\\LargeTestVideo.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
 "bytes": 11531882761,
 "last_write_time": "2026-05-04T17:18:56.0000000+05:30",
- "md5": "25b29aa56ba69b7e34e35d811309bba2"
+ "md5": "25b29aa56ba69b7e34e35d811309bba2",
+ "md5_source": "Get-FileHash"
 },
 {
 "video_id": "gbaaddy",
 "filename": "gBaaddy20May2026.MP4",
 "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled\\gBaaddy20May2026.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
 "bytes": 7075098663,
 "last_write_time": "2026-05-22T08:40:40.0000000+05:30",
- "md5": "d50da42b59dae2fdeec03e47f3b35536"
+ "md5": "d50da42b59dae2fdeec03e47f3b35536",
+ "md5_source": "Get-FileHash"
 },
 {
 "video_id": "testlarge_short",
 "filename": "TestLargeVideo.MP4",
 "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled\\TestLargeVideo.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
 "bytes": 1510427943,
 "last_write_time": "2026-05-04T20:39:14.0000000+05:30",
- "md5": "8fe6b719603dd7a7b2b6ef543139a4a3"
+ "md5": "8fe6b719603dd7a7b2b6ef543139a4a3",
+ "md5_source": "Get-FileHash"
 },
 {
 "video_id": "mahadevpura_singles",
 "filename": "MahadevpuraSingles.MP4",
 "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded
Videos\\Training\\Golden Labelled\\MahadevpuraSingles.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
 "bytes": 196182380,
 "last_write_time": "2026-01-16T08:00:28.0000000+05:30",
- "md5": "915473794e0909c9f95a01e4511a5e98"
+ "md5": "915473794e0909c9f95a01e4511a5e98",
+ "md5_source": "Get-FileHash"
 },
+ {
+ "video_id": "mahadevpura_2",
+ "filename": "mahadevpura-2_proxy.mp4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 3 - Badminton Multicourt\\mahadevpura-2_proxy.mp4",
+ "source_role": "annotation_proxy",
+ "source_status": "present_local",
+ "bytes": 3092000570,
+ "last_write_time": "2026-06-15T15:30:33.2760000+05:30",

```

```

+ "md5": "d75f66340a44cac37357cce9ab988724",
+ "md5_source": "Get-FileHash"
+ },
+ {
+ "video_id": "GX010128",
+ "filename": "GX010128.MP4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 13 - Badminton\\GX010128.MP4",
+ "source_role": "annotation_source",
+ "source_status": "present_local",
+ "bytes": 2342749344,
+ "last_write_time": "2026-06-16T00:25:44.0000000+05:30",
+ "md5": "db84f3e65318e49aea8320de4c44f7d1",
+ "md5_source": "Get-FileHash"
+ },
+ {
+ "video_id": "mahadevpura_1",
+ "filename": "mahadevpura-1.MP4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 14 - Badminton\\mahadevpura-1.MP4",
+ "source_role": "annotation_source",
+ "source_status": "present_local",
+ "bytes": 548827049,
+ "last_write_time": "2026-06-16T00:22:14.0000000+05:30",
+ "md5": "38bef75d78cbbd8209a80fd73c28e6ed",
+ "md5_source": "Get-FileHash"
+ },
+ {
+ "video_id": "GX030094",
+ "filename": "GX030094_proxy.mp4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 6 - Badminton\\GX030094_proxy.mp4",
+ "source_role": "annotation_proxy",
+ "source_status": "present_local",
+ "bytes": 1429944539,
+ "last_write_time": "2026-06-15T17:13:40.5290000+05:30",
+ "md5": "27f1c707bbfb9b81d3b7c6b5afdda7e8",
+ "md5_source": "Get-FileHash"
+ },
+ {
+ "video_id": "Badminton_BXH_2",
+ "filename": "Badminton BXH 2_proxy.mp4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 10 - Badminton Multicourt\\Badminton BXH 2_proxy.mp4",
+ "source_role": "annotation_proxy",
+ "source_status": "present_local",
+ "bytes": 3660841332,
+ "last_write_time": "2026-06-15T16:00:42.7790000+05:30",
+ "md5": "ebebccb702c4be5f3a7da7e000b0f3a4",
+ "md5_source": "Get-FileHash"
+ },
+ {
+ "video_id": "GX010141",
+ "filename": "GX010141.MP4",
+ "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded Videos\\Badminton -
Boxhill - 17 June\\GX010141.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
+ "bytes": 5772876373,
+ "last_write_time": "2026-06-17T09:11:06.0000000+05:30",
+ "md5": "219a84a4e1a985edf5fb039142c18b8d",
+ "md5_source": "Get-FileHash",
+ "notes": "F-drive original found after the drive was connected on 2026-07-04. Earlier
Boxhill run log records derivative GX010141_1080p.mp4 with MD5 daaf8af198640516d14b217f50445d99;
this entry records the present original source file."

```

```

+ },
+ {
+ "video_id": "GX010137",
+ "filename": "GX010137.MP4",
+ "local_path": "F:\\[Khelsutra] GoPro Hero Black 12\\GoPro Recorded Videos\\Badminton -
Boxhill - 17 June\\GX010137.MP4",
+ "source_role": "owned_original",
+ "source_status": "present_local",
+ "bytes": 6762153261,
+ "last_write_time": "2026-06-17T08:21:58.0000000+05:30",
+ "md5": "3d19ef835a69be93acdd75ef636f345a",
+ "md5_source": "Get-FileHash",
+ "notes": "F-drive original found after the drive was connected on 2026-07-04. Earlier
Boxhill run log records derivative GX010137_1080p.mp4 with MD5 74eedbc35d739c231344b8929df4b4a7;
this entry records the present original source file."
+ },
+ {
+ "video_id": "Boxhill_Doubles",
+ "filename": "Boxhill Doubles_proxy.mp4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 12 - Badminton Multicourt\\Boxhill Doubles_proxy.mp4",
+ "source_role": "annotation_proxy",
+ "source_status": "present_local",
+ "bytes": 1553998715,
+ "last_write_time": "2026-06-15T16:23:18.9720000+05:30",
+ "md5": "9a35fffd9bdee397575ffda9d68ebca3",
+ "md5_source": "Get-FileHash"
+ },
+ {
+ "video_id": "GX020094",
+ "filename": "GX020094_proxy.mp4",
+ "local_path": "G:\\My Drive\\Sharing\\Annotation Videos\\Golden Label Videos Request -
June 15\\[Done] Video 4 - Badminton\\GX020094_proxy.mp4",
+ "source_role": "annotation_proxy",
+ "source_status": "present_local",
+ "bytes": 1554122102,
+ "last_write_time": "2026-06-15T16:54:20.8510000+05:30",
+ "md5": "08471270f285123021ef4f9d21d1b05b",
+ "md5_source": "Get-FileHash"
+ }
]
}
diff --git a/tests/test_golden_source_videos.py b/tests/test_golden_source_videos.py
index 8e757e9..cc6fcdd 100644
--- a/tests/test_golden_source_videos.py
+++ b/tests/test_golden_source_videos.py
@@ -14,31 +14,81 @@
 "gbaaddy",
 "testlarge_short",
 "mahadevpura_singles",
+ "mahadevpura_2",
+ "GX010128",
+ "mahadevpura_1",
+ "GX030094",
+ "Badminton_BXH_2",
+ "GX010141",
+ "GX010137",
+ "Boxhill_Doubles",
+ "GX020094",
+ }
+ORIGINAL_SIX_VIDEO_IDS = {
+ "adarsh_avi_singles",
+ "kushagra_singles",
+ "largetest_doubles",
+ "gbaaddy",

```

```

+ "testlarge_short",
+ "mahadevpura_singles",
+ }
+ RUN_LOG_DERIVATIVE_NOTE_VIDEO_IDS = {"GX010137", "GX010141"}

-def test_golden_source_video_manifest_matches_registry_paths() -> None:
+def test_golden_source_video_manifest_covers_indexer_n15_registry() -> None:
 manifest = json.loads(MANIFEST_PATH.read_text(encoding="utf-8"))

- assert manifest["schema"] == "sports-data-collector.golden_source_videos.v1"
+ assert manifest["schema"] == "sports-data-collector.golden_source_videos.v2"
 assert manifest["upload_status"] == "not_uploaded_owner_gated"
+ assert manifest["eval_manifest"] == (
+ "badminton-highlight-indexer/output/human_lovo_manifest.json"
+)

 entries = manifest["videos"]
 assert {entry["video_id"] for entry in entries} == EXPECTED_VIDEO_IDS
+ assert len({entry["md5"] for entry in entries}) == len(entries)
+
+ for entry in entries:
+ assert re.fullmatch(r"[0-9a-f]{32}", entry["md5"])
+ assert entry["source_role"] in {
+ "owned_original",
+ "annotation_source",
+ "annotation_proxy",
+ "boxhill_run_input",
+ }
+
+ assert entry["source_status"] == "present_local"
+ assert entry["local_path"]
+ assert entry["bytes"] > 0
+ assert entry["last_write_time"]
+ assert entry["md5_source"] == "Get-FileHash"
+
+ if entry["video_id"] in RUN_LOG_DERIVATIVE_NOTE_VIDEO_IDS:
+ assert "F-drive original found" in entry["notes"]
+ assert "_1080p.mp4" in entry["notes"]
+
+def test_original_six_manifest_paths_match_collector_db() -> None:
+ manifest = json.loads(MANIFEST_PATH.read_text(encoding="utf-8"))
+ entries = {
+ entry["video_id"]: entry
+ for entry in manifest["videos"]
+ if entry["video_id"] in ORIGINAL_SIX_VIDEO_IDS
+ }
+ assert set(entries) == ORIGINAL_SIX_VIDEO_IDS

 with sqlite3.connect(DB_PATH) as conn:
 rows = {
 video_id: local_path
 for video_id, local_path in conn.execute(
 "SELECT video_id, local_path FROM videos WHERE video_id IN "
- f"({'.'.join('? ' for _ in EXPECTED_VIDEO_IDS)})",
- tuple(EXPECTED_VIDEO_IDS),
+ f"({'.'.join('? ' for _ in ORIGINAL_SIX_VIDEO_IDS)})",
+ tuple(ORIGINAL_SIX_VIDEO_IDS),
)
 }

- assert set(rows) == EXPECTED_VIDEO_IDS
- for entry in entries:
+ assert set(rows) == ORIGINAL_SIX_VIDEO_IDS

```

```

+ source_root = manifest["source_roots"]["original_six"]
+ for entry in entries.values():
+ assert entry["local_path"] == rows[entry["video_id"]]
- assert entry["local_path"].startswith(manifest["source_root"])
- assert re.fullmatch(r"[0-9a-f]{32}", entry["md5"])
- assert entry["bytes"] > 0
+ assert entry["local_path"].startswith(source_root)

```

#### 4. assistant (2026-07-04T00:51:18.965Z)

Now let me extract the 15 video\_ids from the PR-branch golden\_source\_videos.json and compare with the eval manifest names:

#### 5. user (2026-07-04T00:51:19.707Z)

```

1 [
2 {
3 "name": "adarsh_avi_singles",
4 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
5 "fps": 59.94,
6 "frame_width": 1920,
7 "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh
and Avi.rallies.csv"
8 },
9 {
10 "name": "kushagra_singles",
11 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv",
12 "fps": 59.94,
13 "frame_width": 1920,
14 "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv"
15 },
16 {
17 "name": "largetest_doubles",
18 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv",
19 "fps": 29.97,
20 "frame_width": 1920,
21 "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv"
22 },
23 {
24 "name": "gbaaddy",
25 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv",
26 "fps": 59.94,
27 "frame_width": 1920,
28 "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/gBaaddy20May2026.rallies.csv"
29 },
30 {
31 "name": "testlarge_short",
32 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv",
33 "fps": 29.97,
34 "frame_width": 1920,
35 "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/TestLargeVideo.rallies.csv"
36 },
37 {
38 "name": "mahadevpura_singles",
39 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv",

```

```

40 "fps": 29.97,
41 "frame_width": 1920,
42 "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/MahadevpuraSingles.rallies.csv"
43 },
44 {
45 "name": "mahadevpura_2",
46 "fps": 59.94,
47 "frame_width": 1920.0,
48 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_2.rallies.csv",
49 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura2_june12_traj.csv"
50 },
51 {
52 "name": "GX010128",
53 "fps": 59.94,
54 "frame_width": 1920.0,
55 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010128.rallies.csv",
56 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010128_june12_traj.csv"
57 },
58 {
59 "name": "mahadevpura_1",
60 "fps": 59.94,
61 "frame_width": 1920.0,
62 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_1.rallies.csv",
63 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura1_june12_traj.csv"
64 },
65 {
66 "name": "GX030094",
67 "fps": 29.97,
68 "frame_width": 1920.0,
69 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX030094.rallies.csv",
70 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX030094_june15_traj.csv"
71 },
72 {
73 "name": "Badminton_BXH_2",
74 "fps": 59.94,
75 "frame_width": 1920.0,
76 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Badminton_BXH_2.rallies.csv",
77 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/Badminton_BXH_2_june15_traj.csv"
78 },
79 {
80 "name": "GX010141",
81 "fps": 59.94,
82 "frame_width": 1920.0,
83 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010141.rallies.csv",
84 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010141_june17_traj.csv"
85 },
86 {
87 "name": "GX010137",
88 "fps": 59.94,
89 "frame_width": 1920.0,
90 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010137.rallies.csv",

```



```

91 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010137_june17_traj.csv"
92 },
93 {
94 "name": "Boxhill_Doubles",
95 "fps": 29.97,
96 "frame_width": 1920.0,
97 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Boxhill_Doubles.rallies.csv",
98 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/Boxhill_Doubles_june18_traj.csv"
99 },
100 {
101 "name": "GX020094",
102 "fps": 29.97,
103 "frame_width": 1920.0,
104 "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX020094.rallies.csv",
105 "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX020094_june18_traj.csv"
106 }
107]

```

## 6. assistant (2026-07-04T00:51:22.225Z)

Perfect. Now let me verify the exact video\_ids from the PR diff and compare them:

## 7. user (2026-07-04T00:51:25.914Z)

```

PR video_ids count: 15
Eval names count: 15
PR unique count: 15
Eval unique count: 15

```

```

Exact match: True
In PR but not in eval: set()
In eval but not in PR: set()

```

## 8. assistant (2026-07-04T00:51:27.862Z)

Good. Now let me check the badminton-highlight-indexer PR #458 to understand how the consumer uses this manifest:

## 9. user (2026-07-04T00:51:29.320Z)

unknown flag: --no-color

Usage: gh pr diff [<number> | <url> | <branch>] [flags]

Flags:

|                        |                                                                |
|------------------------|----------------------------------------------------------------|
| --color string         | Use color in diff output: {always never auto} (default "auto") |
| -e, --exclude patterns | Exclude files matching glob patterns from the diff             |
| --name-only            | Display only names of changed files                            |
| --patch                | Display diff in patch format                                   |
| -w, --web              | Open the pull request diff in the browser                      |

## 10. user (2026-07-04T00:51:33.576Z)

<persisted-output>

Output too large (51.9KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\tool-results\b1hfhs3t3.txt

Preview (first 2KB):

```
diff --git a/SESSION_HANDOFF.md b/SESSION_HANDOFF.md
index 31ea4c14..a50cc8a1 100644
--- a/SESSION_HANDOFF.md
+++ b/SESSION_HANDOFF.md
@@ -2,49 +2,51 @@
```

## You are here

-A6 is implemented and open for review as PR #455:  
-`https://github.com/Khelsutra/badminton-highlight-indexer/pull/455`.  
+A6 PR #455 is merged. This checkout is on `codex/a7-a9-quality-chain`, fast-forwarded to  
+`origin/master` at `c5fc27c`, with tracked edits in:

-The branch is `codex/a6-ablation-cli`, rebased onto `origin/master` after A5  
-merged as `c8eec7c` / PR #454. It fixes the `python -m backend.eval.ablation`  
-circular import by making `ablation.export\_pdf` a lazy wrapper, adds a  
-module-entrypoint regression test, and records the n=15 R2/tolF1 contrast in  
-`docs/ALPHA\_LAUNCH\_READINESS.md`.  
+- `backend/eval/fusion\_golden.py`  
+- `backend/eval/fusion\_audio.py`  
+- `tests/test\_fusion\_compare.py`  
+- `tests/test\_audio\_features.py`  
+- `docs/ALPHA\_LAUNCH\_READINESS.md`  
+- `SESSION\_HANDOFF.md`

-Validation completed locally on 2026-07-03:  
-`ruff check .`, `python -m mypy backend training`, and  
-`python run\_all\_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70`  
-all passed. Full suite result: 1609 passed, 6 skipped, 85.65% coverage.  
+A7/A8/A9 have been run locally against the 15-video corpus, and draft PR #458 is open:  
+`https://github.com/Khelsutra/badminton-highlight-indexer/pull/458`. The generated  
+feature/model artifacts live under ignored `output/` and `artifacts/` paths and should not  
+be committed.

## Next steps / open threads

-1. Review/merge PR #455 for A6.  
-2. A7 is the next indexer-owned quality item: regenerate or upconvert  
- full-fusion feature JSONs for the newer nine golden videos so fusion/person  
- /audio analyses can run on all 15 videos. GPU is approved on this machine.  
-3. A8 follows A7: rerun `fusion\_compare`, group ablation, and served contrast;  
- explicitly promote or reject fusion/person/Gate-A served-default decisions.  
...  
</persisted-output>

## 11. assistant (2026-07-04T00:51:35.425Z)

Let me read the file directly:

## 12. user (2026-07-04T00:51:36.660Z)

./backend/eval/fusion\_golden.py

## 13. user (2026-07-04T00:51:38.275Z)

```
1 """Offline golden-set person-feature extractor ? the fusion-P3 measurement's GPU step.
2
3 Mirrors `distill_local`'s CSV-driven flow but ADDS the ~5 person-cue features per
shuttle
4 candidate window, by running `PersonDetector` over the ORIGINAL video. It reuses the
5 EXISTING golden trajectory CSVs (so **no WASB re-run**), needs **no Gemini and no DB**,
and
6 writes one small JSON per video. The LOVO comparison (`fusion_compare.py`) then runs
100% on
```

```

7 CPU offline. So the GPU touches exactly one thing ? person detection ? and the lift is
8 measured by the shipped experiment harness; person features enter only as feature
columns the
9 learned scorer weights (no special-casing, default-OFF until the golden ?F1 justifies
them).
10
11 Run (on the GPU box):
12 python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json
--out-dir output
13
14 Each output `output/<name>_golden_features.json` is the replayable feature substrate ?
re-run
15 `fusion_compare` with different variants/thresholds forever without re-detecting on the
GPU.
16 """
17
18 from __future__ import annotations
19
20 import argparse
21 import json
22 import os
23 import time
24 from typing import Any, Dict, List, Optional, Sequence
25
26 from backend.eval import distill_local
27 from backend.eval.calibrate_wasb import load_golden_gts
28 from backend.eval.golden_manifest import load_resolved_manifest
29 from backend.eval.training import label_candidates
30 from backend.pipeline.detectors import fusion_features as ff
31 from backend.pipeline.detectors import rally_gate
32 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
33 from backend.utils.run_telemetry import RunTelemetry
34
35 # The 5 fps/resolution-invariant shuttle features distill_local already computes.
36 SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
37 # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
38 PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
39 SOURCE_MANIFEST_ENV = "RALLY_GOLDEN_SOURCE_MANIFEST"
40
41
42 def is_full_fusion_feature_file(path: str) -> bool:
43 """True when an existing golden feature JSON already has shuttle+person cue dicts.
44
45 Corpus-growth runs can leave leaner windowing-only JSONs with the same filename
pattern.
46 Those are useful for gts/windowing analysis, but they are not the replayable fusion
47 substrate that ``fusion_compare`` and ``ablation`` need.
48 """
49
50 try:
51 with open(path, encoding="utf-8") as f:
52 data = json.load(f)
53 except (OSError, json.JSONDecodeError, TypeError):
54 return False
55 candidates = data.get("candidates")
56 if not isinstance(candidates, list):
57 return False
58 if not candidates:
59 return True
60 return all(
61 isinstance(candidate, dict)
62 and "shuttle" in candidate
63 and "person" in candidate
64 for candidate in candidates
65)

```

```

66
67
68 def load_source_video_paths(source_manifest_path: Optional[str]) -> Dict[str, str]:
69 """Load ``video_id -> local_path`` from the collector golden-source manifest.
70
71 The collector records source/proxy ownership and full-file MD5s. Only entries marked
72 ``present_local`` with a concrete ``local_path`` can drive local GPU extraction.
73 """
74
75 if not source_manifest_path:
76 return {}
77 with open(source_manifest_path, encoding="utf-8") as f:
78 data = json.load(f)
79 entries = data.get("videos", data) if isinstance(data, dict) else data
80 if not isinstance(entries, list):
81 raise ValueError(f"source manifest must contain a videos list:
{source_manifest_path}")
82
83 paths: Dict[str, str] = {}
84 for entry in entries:
85 if not isinstance(entry, dict):
86 continue
87 video_id = entry.get("video_id") or entry.get("name")
88 local_path = entry.get("local_path")
89 if not video_id or not local_path:
90 continue
91 if entry.get("source_status", "present_local") != "present_local":
92 continue
93 paths[str(video_id)] = str(local_path)
94 return paths
95
96
97 def attach_source_video_paths(
98 specs: Sequence[Dict[str, Any]], source_manifest_path: Optional[str]
99) -> int:
100 """Attach explicit ``video_path`` fields to manifest specs from the source manifest.
101
102 Returns the number of specs updated. The collector source manifest is authoritative
103 for local source/proxy video location and overrides derived/stale manifest paths.
104 """
105
106 source_paths = load_source_video_paths(source_manifest_path)
107 updated = 0
108 for spec in specs:
109 name = str(spec.get("name", ""))
110 source_path = source_paths.get(name)
111 if not source_path:
112 continue
113 if spec.get("video_path") == source_path:
114 continue
115 spec["video_path"] = source_path
116 updated += 1
117 return updated
118
119
120 def _filter_specs(
121 specs: Sequence[Dict[str, Any]], only: Sequence[str]
122) -> List[Dict[str, Any]]:
123 if not only:
124 return list(specs)
125 wanted = list(dict.fromkeys(only))
126 wanted_set = set(wanted)
127 filtered = [spec for spec in specs if str(spec.get("name")) in wanted_set]
128 found = {str(spec.get("name")) for spec in filtered}
129 missing = [name for name in wanted if name not in found]

```

```

130 if missing:
131 raise ValueError(f"--only requested unknown video(s): {'', '.join(missing)}")
132 return filtered
133
134
135 def _derive_video_path(spec: Dict[str, Any]) -> str:
136 """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
137 golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming
138 convention)."""
139 if spec.get("video_path"):
140 return str(spec["video_path"])
141 labels = str(spec["labels"])
142 if labels.endswith(".rallies.csv"):
143 return labels[: -len(".rallies.csv")] + ".MP4"
144 raise ValueError(
145 f"{spec.get('name')}: no video_path and labels {labels!r} isn't *.rallies.csv"
146)
147
148 def extract_video(
149 spec: Dict[str, Any],
150 detector: Optional[Any] = None,
151 frame_skip: int = 3,
152 iou: float = 0.5,
153 log_every_sec: float = 15.0,
154 telemetry: Optional[Any] = None,
155 use_gpu: bool = True,
156 detector_model: Optional[str] = None,
157 max_gpu_temp_c: Optional[float] = None,
158) -> Dict[str, Any]:
159 """One golden video -> a feature record (shuttle + person features per candidate
160 window).
161
162 Returns `{name, fps, frame_width,
163 candidates:[{start,end,shuttle{ },person{ },label}], gts}``.
164 `detector` (anything with `detections_over_windows`) is injectable for GPU-free
165 tests;
166 when None a real `PersonDetector` is built (loads torchvision on first use).
167 `telemetry` (a `RunTelemetry` or None) records per-frame velocity + machine-health
168 snapshots
169 from the detection progress callback so a hard kill leaves a forensic runlog.
170 """
171 fps, fw = float(spec["fps"]), float(spec["frame_width"])
172 traj = str(spec["trajectory"])
173 points = TrackNetRunner.parse_trajectory_csv(traj)
174 intervals, x_shuttle = distill_local.build_candidates(traj, fps, fw)
175 gts = load_golden_gts(str(spec["labels"]))
176 labels = label_candidates(intervals, gts, iou)
177 # Net axis for the person two-sided split ? reuse rally_gate's camera-agnostic
178 estimate
179 # (parity with the shuttle net-crossing gate) rather than hard-coding an
180 orientation.
181 axis, net_px, _span = rally_gate.estimate_play_axis(points)
182 video_path = _derive_video_path(spec)
183 if not os.path.isfile(video_path):
184 raise FileNotFoundError(
185 f"{spec.get('name')}: source video not found for fusion extraction:
186 {video_path}"
187)
188
189 if detector is None:
190 from backend.pipeline.detectors.person_detector import PersonDetector
191
192 indexing_cfg: Dict[str, Any] = {"use_gpu": use_gpu}
193 if detector_model:

```

```

187 indexing_cfg["detector_model"] = detector_model
188 detector = PersonDetector({"indexing": indexing_cfg})
189
190 # Single-pass person detection over ALL candidate windows (open the video ONCE, read
191 # forward ? no per-window seek). Then compute features per window (pure, no GPU).
192 n = len(intervals)
193 win_frames = [(round(s * fps), round(e * fps)) for (s, e) in intervals]
194 print(
195 f"[fusion_golden] {spec['name']}: {n} windows ? person-detecting in one
pass...",
196 flush=True,
197)
198 t0 = time.monotonic()
199 state = {"last": t0}
200
201 def _progress(done: int, total: int) -> None:
202 rec = None
203 if telemetry is not None:
204 rec = telemetry.snapshot(done, total, phase=spec["name"])
205 if max_gpu_temp_c is not None and rec is not None:
206 gpu = rec.get("gpu") or {}
207 temp = gpu.get("temp_c")
208 if temp is not None and float(temp) >= max_gpu_temp_c:
209 raise RuntimeError(
210 f"{spec['name']}: GPU temperature {float(temp):.1f}C reached "
211 f"--max-gpu-temp-c {max_gpu_temp_c:.1f}; aborting before thermal
shutdown"
212)
213 now = time.monotonic()
214 if now - state["last"] >= log_every_sec or done >= total:
215 el = now - t0
216 rate = done / el if el > 0 else 0.0
217 eta = (total - done) / rate if rate > 0 else 0.0
218 pct = 100 * done // total if total else 0
219 print(
220 f"[fusion_golden] {spec['name']}: detect frame {done}/{total} ({pct}%)
"
221 f"| {el:.0f}s elapsed | ETA {eta:.0f}s",
222 flush=True,
223)
224 state["last"] = now
225
226 per_window = detector.detections_over_windows(
227 video_path, win_frames, frame_skip, progress_cb=_progress
228)
229
230 candidates: List[Dict[str, Any]] = []
231 for i, ((s, e), xs) in enumerate(zip(intervals, x_shuttle)):
232 det = per_window[i]
233 det_fw = det.get("frame_width") or fw
234 det_fh = det.get("frame_height") or 0.0
235 # Normalise the (pixel) net coord to 0-1 on the play axis, matching the foot-
point
236 # normalisation fusion_features uses (x/width, y/height).
237 if axis == "x":
238 net_norm = net_px / det_fw if det_fw > 0 else 0.5
239 else:
240 net_norm = net_px / det_fh if det_fh > 0 else 0.5
241 sf, ef = win_frames[i]
242 shuttle_in = [p for p in points if sf <= p.frame <= ef]
243 person = ff.compute_person_features(
244 det["frames"],
245 shuttle_in,
246 det_fw,
247 det_fh,

```

```

248 court_polygon=None,
249 net=(axis, net_norm),
250)
251 candidates.append(
252 {
253 "start": float(s),
254 "end": float(e),
255 "shuttle": {k: float(v) for k, v in zip(SHUTTLE_FEATURE_NAMES, xs)},
256 "person": {k: float(v) for k, v in person.items()},
257 "label": int(labels[i]),
258 }
259)
260 return {
261 "name": spec["name"],
262 "fps": fps,
263 "frame_width": fw,
264 "candidates": candidates,
265 "gts": [[float(a), float(b)] for a, b in gts],
266 }
267
268
269 def run(
270 manifest_path: str,
271 out_dir: str,
272 frame_skip: int = 3,
273 iou: float = 0.5,
274 fresh: bool = False,
275 smallest_first: bool = False,
276 source_manifest: Optional[str] = None,
277 only: Sequence[str] = (),
278 missing_fusion_only: bool = False,
279 use_gpu: bool = True,
280 detector_model: Optional[str] = None,
281 max_gpu_temp_c: Optional[float] = None,
282) -> List[str]:
283 specs = load_resolved_manifest(manifest_path)
284 source_manifest = source_manifest or os.environ.get(SOURCE_MANIFEST_ENV)
285 if source_manifest:
286 updated = attach_source_video_paths(specs, source_manifest)
287 print(
288 f"[fusion_golden] attached {updated}/{len(specs)} source video path(s) "
289 f"from {source_manifest}",
290 flush=True,
291)
292 specs = _filter_specs(specs, only)
293 if smallest_first:
294 # Process small videos first for quick feedback (the slow full match lands
295 # last);
296 # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
297 # row/frame).
298 specs = sorted(
299 specs,
300 key=lambda s: (
301 os.path.getsize(s["trajectory"])
302 if os.path.isfile(s.get("trajectory", ""))
303 else 0
304),
305)
306 os.makedirs(out_dir, exist_ok=True)
307 # Run-telemetry black box: a long GPU run can be OS-killed with no traceback, so
308 # keep a tiny
309 # rotating runlog of velocity + machine health for the post-mortem. Constructing it
310 # must never
311 # break extraction, so guard the build and fall back to no telemetry on any error.
312 telem: Optional[RunTelemetry]

```

```

309 try:
310 telem = RunTelemetry(
311 os.path.join(out_dir, "run_telemetry.jsonl"), label="fusion_golden"
312)
313 except Exception: # noqa: BLE001 - telemetry wiring must never break the run it
observes
314 telem = None
315 written: List[str] = []
316 for vi, spec in enumerate(specs, 1):
317 out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
318 existing = os.path.isfile(out)
319 existing_full = existing and is_full_fusion_feature_file(out)
320 # Resumable: skip a video whose full-fusion JSON already exists. A leaner
321 # windowing-only JSON is stale for A7, so re-extract it even without --fresh.
322 if missing_fusion_only and not fresh and existing_full:
323 print(
324 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: full-fusion "
325 f"features already present -> {out}",
326 flush=True,
327)
328 written.append(out)
329 continue
330 if not fresh and existing and existing_full:
331 print(
332 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already extracted
333 "
334 f"-> {out} (use --fresh to redo)",
335 flush=True,
336)
337 written.append(out)
338 continue
339 if not fresh and existing and not existing_full:
340 print(
341 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: existing JSON "
342 f"is not full-fusion schema; re-extracting...",
343 flush=True,
344)
345 print(
346 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",
347 flush=True,
348)
349 t0 = time.monotonic()
350 if telem is not None:
351 telem.event("video_start", spec["name"])
352 try:
353 data = extract_video(
354 spec,
355 frame_skip=frame_skip,
356 iou=iou,
357 telemetry=telem,
358 use_gpu=use_gpu,
359 detector_model=detector_model,
360 max_gpu_temp_c=max_gpu_temp_c,
361)
362 except Exception as e: # noqa: BLE001 - record machine state at death, then re-
raise
363 if telem is not None:
364 telem.event("error", spec["name"], err=str(e))
365 raise
366 n_pos = sum(c["label"] for c in data["candidates"])
367 with open(out, "w", encoding="utf-8") as f:
368 json.dump(data, f, indent=2)
369 if telem is not None:
370 telem.event(
371 "video_done",

```



```

371 spec["name"],
372 windows=len(data["candidates"]),
373 positives=n_pos,
374)
375 print(
376 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: DONE in
{time.monotonic() - t0:.0f}s "
377 f"| {len(data['candidates'])} windows ({n_pos} positive) |
{len(data['gts'])} golden rallies "
378 f"-> {out}",
379 flush=True,
380)
381 written.append(out)
382 return written
383
384
385 def main() -> None:
386 ap = argparse.ArgumentParser(
387 description="Extract golden-set shuttle+person features (fusion P3 GPU step)."
388)
389 ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
390 ap.add_argument("--out-dir", default="output")
391 ap.add_argument("--frame-skip", type=int, default=3)
392 ap.add_argument("--iou", type=float, default=0.5)
393 ap.add_argument(
394 "--fresh", action="store_true", help="re-extract even if a JSON already exists"
395)
396 ap.add_argument(
397 "--smallest-first",
398 action="store_true",
399 help="process smaller videos first for quick feedback (LOVO is order-
independent)",
400)
401 ap.add_argument(
402 "--source-manifest",
403 default=None,
404 help=(
405 "collector golden_source_videos.json; overrides video paths for source/proxy
"
406 f"videos (or set {SOURCE_MANIFEST_ENV})"
407),
408)
409 ap.add_argument(
410 "--only",
411 action="append",
412 default=[],
413 metavar="VIDEO_ID",
414 help="process one manifest video by name; repeatable",
415)
416 ap.add_argument(
417 "--missing-fusion-only",
418 action="store_true",
419 help="skip videos whose existing JSON already has full shuttle+person fusion
schema",
420)
421 ap.add_argument(
422 "--cpu",
423 action="store_true",
424 help="run torchvision person detection on CPU instead of CUDA",
425)
426 ap.add_argument(
427 "--detector-model",
428 default=None,
429 choices=[
430 "fasterrcnn_resnet50_fpn",

```

```

431 "fasterrcnn_mobilenet_v3_large_fpn",
432 "retinanet_resnet50_fpn",
433],
434 help="torchvision detector model; default keeps the configured Faster R-CNN
ResNet",
435)
436 ap.add_argument(
437 "--max-gpu-temp-c",
438 type=float,
439 default=None,
440 help="abort extraction if telemetry sees GPU temperature at or above this
value",
441)
442 args = ap.parse_args()
443 run(
444 args.manifest,
445 args.out_dir,
446 args.frame_skip,
447 args.iou,
448 fresh=args.fresh,
449 smallest_first=args.smallest_first,
450 source_manifest=args.source_manifest,
451 only=args.only,
452 missing_fusion_only=args.missing_fusion_only,
453 use_gpu=not args.cpu,
454 detector_model=args.detector_model,
455 max_gpu_temp_c=args.max_gpu_temp_c,
456)
457
458
459 if __name__ == "__main__":
460 main()
461

```

#### 14. assistant (2026-07-04T00:51:41.233Z)

Now let me check the ALPHA\_LAUNCH\_READINESS.md document in PR #458:

#### 15. user (2026-07-04T00:51:43.645Z)

<persisted-output>

Output too large (31.4KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\tool-results\b9e7fgawr.txt

Preview (first 2KB):

```

- `docs/ALPHA_LAUNCH_READINESS.md`.
+- `backend/eval/fusion_golden.py`
+- `backend/eval/fusion_audio.py`
+- `tests/test_fusion_compare.py`
+- `tests/test_audio_features.py`
+- `docs/ALPHA_LAUNCH_READINESS.md`
+- `SESSION_HANDOFF.md`

```

-Validation completed locally on 2026-07-03:

```

- `ruff check .`, `python -m mypy backend training`, and
- `python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70`
-all passed. Full suite result: 1609 passed, 6 skipped, 85.65% coverage.
+A7/A8/A9 have been run locally against the 15-video corpus, and draft PR #458 is open:
+`https://github.com/Khelsutra/badminton-highlight-indexer/pull/458`. The generated
+feature/model artifacts live under ignored `output/` and `artifacts/` paths and should not
+be committed.

```

## Next steps / open threads

- 1. Review/merge PR #455 for A6.
- 2. A7 is the next indexer-owned quality item: regenerate or upconvert
  - full-fusion feature JSONs for the newer nine golden videos so fusion/person
  - /audio analyses can run on all 15 videos. GPU is approved on this machine.
- 3. A8 follows A7: rerun `fusion\_compare`, group ablation, and served contrast;
  - explicitly promote or reject fusion/person/Gate-A served-default decisions.
- 4. A3 is owned by another agent; do not duplicate that collector/vault work.

- +1. Address review comments on draft PR #458; convert to ready only after owner/reviewer agrees
  - + the A9 TemporalMaxer scope question is settled.
- +2. Decide whether A9 is considered complete with the current Gen-0 non-serving shadow
  - + bundle or whether TemporalMaxer needs a follow-up scorer-swap PR.
- +3. Do not touch A1; another agent owns it.
- +4. A3 is in collector PR #62 and records all 15 source videos; the two Boxhill originals
  - + were verified on `F:\`.

## Ramp-up kit

```
- `docs/ALPHA_LAUNCH_READINESS.md`
-- `docs/data_pipeline/GOLDEN_VIDEOS.md`
-- `docs/R2_EVAL_METRIC_DESIGN.md`
-- `backend/eval/ablation.py`
-- `backend/eval/ablation_pdf.py`
...
</persisted-output>
```

## 16. assistant (2026-07-04T00:51:45.570Z)

Let me read the ALPHA\_LAUNCH\_READINESS.md file directly:

## 17. user (2026-07-04T00:51:45.882Z)

```
1 # Alpha Launch Readiness Plan
2
3 > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-04
4 > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5 > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6 > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheIsutra` Studio
docs.
7 > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9 ---
10
11 ## 0. TL;DR
12
13 The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` A2-A7 made the local manifest/path and full-
fusion/audio feature surface reproducible on all 15; `[analysis]` A4/A5 refreshed the n=15
nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default promotion
and owned-model serving claims on the current evidence.
14
15 The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17 ## 1. Problem & goal
18
19 The previous launch blockers were spread across roadmap docs and mixed together three
```

different questions:

20

- 21 - **\*\*Can an invited tester use the product without SSH?\*\***
- 22 - **\*\*Can the 15-video golden corpus now support honest quality gates?\*\***
- 23 - **\*\*Can we claim or ship an owned/commercial rally detector?\*\***

24

25 This doc separates those into executable workstreams. The goal is to reach an invite-only alpha where the product is usable end-to-end, the quality claims are honest and reproducible, and the remaining model/legal gates are explicit rather than accidentally implied by launch activity.

26

27 ## 2. Decisions locked

28

29 | # | Decision | Source / date | Implication |

30 |---|---|---|---|

31 | D1 | **\*\*Product alpha and owned-model launch are separate decisions.\*\*** |

`NEXT\_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without claiming the owned model is commercially cleared. |

32 | D2 | **\*\*First alpha must be page-driven, not SSH/CLI-driven.\*\*** | `NEXT\_STEPS.md` P0 UI-driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling, compile, and download are launch-critical. |

33 | D3 | **\*\*The 15-video corpus is now a measurement asset, not a free launch pass.\*\*** |   
`[verified]` `docs/data\_pipeline/GOLDEN\_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 |   
Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims. |

34 | D4 | **\*\*Do not promote served Gate A by default from the current n=15 evidence.\*\*** |   
`[analysis]` `backend.eval.serve\_contrast` run, 2026-07-04 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |

35 | D5 | **\*\*P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.\*\*** |   
sibling `khelsutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable. |

36 | D6 | **\*\*Gen-0 n=15 shadow training is non-serving.\*\*** | `[analysis]`   
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired sign test is not significant; #172 and WASB C3 remain hard blockers. |

37

38 ## 3. Verified snapshot, 2026-07-03

39

40 All five Git checkouts were fast-forwarded before analysis:

41

42 | Repo | Branch | Synced HEAD |

43 |---|---|---|

44 | `badminton-highlight-indexer` | `master` | `e4914d1` |

45 | `khelsutra` | `master` | `b0f70a5` |

46 | `rally-annotator` | `main` | `ef9e088` |

47 | `sports-data-collector` | `master` | `bfc014f` |

48 | `sports-obsreport` | `main` | `a0c7252` |

49

50 Corpus/eval facts:

51

52 - `[verified]` `docs/data\_pipeline/GOLDEN\_VIDEOS.md` says   
`output/human\_lovo\_manifest.json` is the ingested golden corpus at `n = 15`.

53 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found all 15 trajectories and labels in this workspace.

54 - `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8 audio augmentation populated audio features for all 15.

55 - `[analysis]` `python -m backend.eval.fusion\_compare --features-dir output` now runs over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so person fusion is not promotable by itself.

56 - `[analysis]` `python -m backend.eval.fusion\_audio --features-dir output --compare` measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive cue, not a default-flip.

57 - `[analysis]` `python -m backend.eval.serve\_contrast --manifest   
output\human\_lovo\_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.

```

58 - `[analysis]` `python -m training.gen0.harness --manifest
output\human_lovo_manifest.json --floor eval_baselines\heuristic_lovo_n15.json --version
0.1.1-n15-shadow.20260704 --out artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`,
floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
59 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
60 - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output
--granularity group` failed at import due to a circular import between
`backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
61
62 ## 4. Launch strategy
63
64 ### 4.1 Alpha lane: make the product usable
65
66 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
67
68 ### 4.2 Quality lane: make the 15-video corpus gate-ready
69
70 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
71
72 ### 4.3 Flywheel lane: make tester feedback useful
73
74 The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
75
76 ## 5. Risks and guardrails
77
78 | Risk | Why it matters | Guardrail |
79 |---|---|---|
80 | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
81 | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
82 | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
83 | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
84 | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |
85 | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |
86
87 ## 6. Honest limits
88
89 Completing this doc's alpha lane does not prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.
90
91 Completing the quality lane does not by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.
92
93 ## 7. Deliverables & progress tracker

```

94

95 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =  
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the  
row explicitly names a cross-repo batch.

96

| ID  | Deliverable                                                                                                                                                                 | Repo / owner                                                   | Depends on       | Gated?                                                         | Status      | PR            |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------|----------------------------------------------------------------|-------------|---------------|
| A0  | Create this alpha readiness tracker and index it in docs                                                                                                                    | `badminton-highlight-indexer`                                  | -                | No                                                             | DONE        | #451          |
| A1  | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified                      | `khelesutra` + `badminton-highlight-indexer`                   | A0               | Owner spend / CF config                                        | TODO        | -             |
| A2  | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing                                | `badminton-highlight-indexer`                                  | A0               | No                                                             | DONE        | #452          |
| A3  | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus                  | `sports-data-collector` + vault                                | A2               | Owner upload/storage scope                                     | IN PROGRESS | collector #62 |
| A4  | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed                                 | `badminton-highlight-indexer`                                  | A2               | Reviewer sign-off                                              | DONE        | #453          |
| A5  | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons                                                             | `badminton-highlight-indexer`                                  | A2               | No                                                             | DONE        | #454          |
| A6  | Fix `backend.eval.ablation` CLI circular import, then run the R2/toIF1 ablation on the n=15 corpus                                                                          | `badminton-highlight-indexer`                                  | A2               | No                                                             | DONE        | #455          |
| A7  | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15                                                    | `badminton-highlight-indexer`                                  | A2               | GPU/data availability                                          | IN PROGRESS | this PR       |
| A8  | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer`                                  | A6, A7           | Reviewer sign-off                                              | IN PROGRESS | this PR       |
| A9  | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict                                            | `badminton-highlight-indexer`                                  | A5, A8           | Owner approval for M0.4 scope                                  | IN PROGRESS | this PR       |
| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed                                       | Owner + `badminton-highlight-indexer`                          | A4, A5, A8       | Owner decision                                                 | GATED       | -             |
| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail                                      | Owner/legal                                                    | A10              | Legal/provenance                                               | GATED       | -             |
| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train`/`reeval` job, delta-F1 surface, and alignment with `COMPUTE_DECOUPLED_SERVING` | M11   `khelesutra` + `badminton-highlight-indexer` + collector | A1, A3           | Owner D13 reopen (`khelesutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED       | -             |
| A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure triage, known limits, rollback/disable steps, and owner GO/NO-GO                              | Cross-repo                                                     | A1, A4, A10, A11 | Owner launch decision                                          | TODO        | -             |

113

114 ## 8. Concrete next PR sequence

115

1. \*\*Docs PR:\*\* land this tracker.
- 117 2. \*\*Corpus portability PR:\*\* rebase/portable manifest tooling plus missing-file smoke test.
- 118 3. \*\*Hosted alpha PR/batch:\*\* health/auth/upload/process/job/compile/download verification and runbook updates.
- 119 4. \*\*Baseline PR:\*\* intentional n=15 nightly baseline refresh and heuristic floor refresh.
- 120 5. \*\*Ablation/fusion PRs:\*\* fix ablation import, complete 15-video features, rerun promotion/rejection decisions.
- 121 6. \*\*Launch decision PR/docs update:\*\* record ship-gate target, C3 status, alpha copy limits, and launch checklist.

122

123 A2 smoke command:

124

125 ```powershell

```

126 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15
127 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15 --write output/human_lovo_manifest.local.json
128 ```
129
130 The checker resolves stale absolute paths against the repo's sibling `Annotation Setup`
folders and `output/`, then exits non-zero if any required `trajectory` or `labels` file is
absent.
131
132 A4 nightly baseline refresh record:
133
134 ```powershell
135 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --update-baseline
136 ```
137
138 Result:
139
140 ```text
141 INFO baseline snapshot written to C:\Users\avidu\Projects\khelsutra-guru\badminton-
highlight-indexer\eval_baselines\nightly_baseline.json
142 INFO default: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
143 INFO milestone: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
144 INFO baseline_off: n=15 tolF1=0.138 f1@0.5=0.237 merge_rate=0.638
145 ```
146
147 Pass-mode proof:
148
149 ```powershell
150 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --no-report
151 ```
152
153 Result: `PASS`, exit code `0`, baseline snapshot `2026-07-03`, default/milestone `tol_f1
= 0.3636`, `f1@0.5 = 0.4723`, `merge_rate = 0.1350`; baseline-off `tol_f1 = 0.1382`, `f1@0.5 =
0.2366`, `merge_rate = 0.6380`.
154
155 A5 heuristic floor refresh record:
156
157 ```powershell
158 python -m backend.eval.calibrate_local --manifest output\human_lovo_manifest.json
--metric f1 --floor-out eval_baselines\heuristic_lovo_n15.json
159 ```
160
161 Result:
162
163 ```text
164 baseline (untuned) mean held-out f1: 0.237
165 calibrated mean held-out f1: 0.446 +/- 0.239 (gap +0.000)
166 ```
167
168 The committed floor is `eval_baselines/heuristic_lovo_n15.json`: `n = 15`, `mean_f1 =
0.4456`, `std = 0.2390`, `metric = f1`, `iou = 0.5`. Every leave-one-video-out fold selected
`(0.05, 1.0, 2.0, 0)`.
169
170 A6 ablation/R2 record:
171
172 ```powershell
173 python -m backend.eval.ablation --features-dir output --granularity group
174 python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
175 ```
176

```

177 Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6` because only the original six feature JSONs carry the full fusion schema; A7 must regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15 R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1 = 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign `14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for the baseline-off to served milestone contrast rather than a blanket fusion/default-flip approval.

178

179 A7 full-fusion/audio feature refresh record:

180

181 ```powershell

182 python -m backend.eval.fusion\_golden --manifest output\human\_lovo\_manifest.json --out-dir output --source-manifest ..\sports-data-collector\data\golden\_source\_videos.json --missing-fusion-only --smallest-first --max-gpu-temp-c 87

183 python -m backend.eval.fusion\_audio --manifest output\human\_lovo\_manifest.json --features-dir output --source-manifest ..\sports-data-collector\data\golden\_source\_videos.json --augment

184 ```

185

186 Result: the GPU pass completed without a thermal abort after fans were set to max performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are present for `15/15`. The large regenerated clips completed as follows: `mahadevpura\_1` `1` window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`, `GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill\_Doubles` `59` / `12` / `43`, `mahadevpura\_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton\_BXH\_2` `29` / `1` / `44`. The source manifest supplied all 15 source-video paths, including the F-drive Boxhill originals.

187

188 A8 fusion/default-decision record:

189

190 ```powershell

191 python -m backend.eval.fusion\_compare --features-dir output

192 python -m backend.eval.fusion\_audio --features-dir output --compare

193 python -m backend.eval.ablation --features-dir output --granularity group

194 python -m backend.eval.serve\_contrast --manifest output\human\_lovo\_manifest.json

195 python -m backend.eval.rally\_seg\_eval --manifest output\human\_lovo\_manifest.json

--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served --vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6\_r2\_ablation.json

196 ```

197

198 Result:

199

200 - Person fusion is \*\*not promoted\*\*: shuttle-only F1 `0.302`, shuttle+person F1 `0.311`, `Delta F1 +0.008`, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.

201 - Audio is \*\*shadow/suggestive, not promoted\*\*: shuttle+person F1 `0.311`, shuttle+person+audio F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`, `p = 0.180`.

202 - Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017, +0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate\_a `-0.001` CI `[-0.012, +0.008]`; person `-0.009` CI `[-0.045, +0.021]`.

203 - Gate A served default is \*\*rejected\*\*: proposal mean `Delta F1 +0.0287`, CI `[-0.0222, +0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.

204 - R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs baseline-off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta tolF1 +0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes, strict per-video split-count guard still blocks blanket promotion language.

205

206 A9 Gen-0 shadow-training record:

207

208 ```powershell

209 python -m training.gen0.harness --manifest output\human\_lovo\_manifest.json --floor eval\_baselines\heuristic\_lovo\_n15.json --out artifacts\rally\_tal\_gen0 --version



```

0.1.1-n15-shadow.20260704
210 ``
211
212 Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_tal_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus
heuristic floor `0.4456`, so `floor_passed=True`. The gate remains **non-serving**: paired sign
test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and
WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the
evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4
scorer-swap PR.
213
214 ## 9. Open questions
215
216 1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
217 2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
218 3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
219 4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
220 5. **Engineering:** Should A9 close on the current Gen-0 non-serving bundle, or should
the actual TemporalMaxer scorer swap be implemented before marking A9 done?
221
222 ## 10. Definition of done
223
224 - [] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
225 - [] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
226 - [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
227 - [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
228 - [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
229 - [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
230 - [] Ship-gate target and allowed alpha claims are recorded.
231 - [] WASB C3 status is recorded and reflected in alpha messaging.
232 - [] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
233 - [] Owner records GO/NO-GO for the first alpha wave.
234
235 ## 11. References
236
237 Internal docs:
238
239 - `docs/NEXT_STEPS.md`
240 - `docs/ROADMAP.md`
241 - `docs/data_pipeline/GOLDEN_VIDEOS.md`
242 - `docs/R2_EVAL_METRIC_DESIGN.md`
243 - `docs/RALLY_DETECTION_FIXES_PLAN.md`
244 - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
245 - sibling `kheIsutra/NEXT_STEPS.md`
246 - sibling `kheIsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
247 - sibling `kheIsutra/docs/PER_RALLY_SELECTION.md`
248 - sibling `sports-data-collector/data/golden_source_videos.json`
249
250 Local verification commands used for the 2026-07-03 snapshot:
251
252 ```powershell
253 python -m backend.eval.fusion_compare --features-dir output
254 python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
255 python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n15.json --version analysis-n15 --out <temp-out>
256 python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir

```

```

output --no-report
257 python -m backend.eval.ablation --features-dir output --granularity group
258 python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
259 ```
260
261 ---
262
263 ### Changelog
264
265 - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
266 - 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default
decisions, and A9 Gen-0 n15 shadow-training gate verdict.
267

```

## 18. assistant (2026-07-04T00:51:48.101Z)

Perfect. Now I have what I need. Let me check a few more details about the actual source\_status values and paths, particularly for the problematic videos:

## 19. user (2026-07-04T00:52:00.163Z)

```

<stdin>:11: SyntaxWarning: invalid escape sequence '\['
<stdin>:18: SyntaxWarning: invalid escape sequence '\['
=== CROSS-REPO CONSISTENCY CHECK ===

```

1. All video source\_status values are: {'present\_local'}
  - > All match filter criteria 'present\_local': TRUE
2. GX010137 note: F-drive original found after the drive was connected on 2026-07-04. Earlier Boxhill run log records derivative GX010137\_1080p.mp4 with MD5 74eedbc35d739c231344b8929df4b4a7; this entry records the present original source file.
  - > Records derivative GX010137\_1080p.mp4 with MD5: 74eedbc35d739c231344b8929df4b4a7
  - > This entry is the ORIGINAL, with different MD5
3. GX010141 note: F-drive original found after the drive was connected on 2026-07-04. Earlier Boxhill run log records derivative GX010141\_1080p.mp4 with MD5 daaf8af198640516d14b217f50445d99; this entry records the present original source file.
  - > Records derivative GX010141\_1080p.mp4 with MD5: daaf8af198640516d14b217f50445d99
  - > This entry is the ORIGINAL, with different MD5
4. fusion\_golden.py load\_source\_video\_paths() logic:
  - Filters on: entry.get('source\_status', 'present\_local') == 'present\_local'
  - Both GX010137 and GX010141 have source\_status='present\_local' ?
  - Both have non-empty local\_path ?
  - Will both be returned in video\_id -> local\_path dict ?
5. fusion\_golden.py attach\_source\_video\_paths() logic:
  - Uses: spec.get('name') as lookup key into source\_paths dict
  - Eval manifest has entries with 'name': 'GX010137', 'name': 'GX010141'
  - Will successfully attach paths for both ?
6. ALPHA\_LAUNCH\_READINESS.md line 186:
 

Claim: "The source manifest supplied all 15 source-video paths, including the F-drive Boxhill originals."

  - GX010137: marked present\_local with F-drive path ?
  - GX010141: marked present\_local with F-drive path ?
  - Claim is CONSISTENT with PR #62 manifest ?
7. POTENTIAL CROSS-REPO BREAK:
  - The eval trajectories/labels in human\_lovo\_manifest.json were created from VIDEO FILES
  - GX010137 note says derivative was GX010137\_1080p.mp4 (MD5: 74eedbc35d739c231344b8929df4b4a7)

- But PR #62 records the ORIGINAL GX010137.MP4 with DIFFERENT MD5
- If eval labels were made from the 1080p derivative, frame indices won't match the original!
- Similarly for GX010141 (derivative MD5: daaf8af198640516d14b217f50445d99)

## 20. assistant (2026-07-04T00:52:02.153Z)

Now let me check the annotation proxy files more carefully:

## 21. user (2026-07-04T00:52:12.358Z)

```
<stdin>:7: SyntaxWarning: invalid escape sequence '\M'
<stdin>:13: SyntaxWarning: invalid escape sequence '\M'
<stdin>:19: SyntaxWarning: invalid escape sequence '\M'
<stdin>:25: SyntaxWarning: invalid escape sequence '\M'
<stdin>:31: SyntaxWarning: invalid escape sequence '\M'
=== ANNOTATION PROXY ANALYSIS ===
```

Five videos are marked as 'annotation\_proxy' (likely what annotations were made from):

```
mahadevpura_2: mahadevpura-2_proxy.mp4
 Path: G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\[Done]
Video 3 - Badminton Multicourt\mahadevpura-2_proxy.mp4
 MD5: d75f66340a44cac37357cce9ab988724
```

```
GX030094: GX030094_proxy.mp4
 Path: G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\[Done]
Video 6 - Badminton\GX030094_proxy.mp4
 MD5: 27f1c707bbfb9b81d3b7c6b5afdda7e8
```

```
Badminton_BXH_2: Badminton BXH 2_proxy.mp4
 Path: G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\[Done]
Video 10 - Badminton Multicourt\Badminton BXH 2_proxy.mp4
 MD5: ebebccb702c4be5f3a7da7e000b0f3a4
```

```
Boxhill_Doubles: Boxhill Doubles_proxy.mp4
 Path: G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\[Done]
Video 12 - Badminton Multicourt\Boxhill Doubles_proxy.mp4
 MD5: 9a35ffffdfbdee397575ffda9d68ebca3
```

```
GX020094: GX020094_proxy.mp4
 Path: G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\[Done]
Video 4 - Badminton\GX020094_proxy.mp4
 MD5: 08471270f285123021ef4f9d21d1b05b
```

fusion\_golden.py behavior when attaching source paths:

- Calls load\_source\_video\_paths() which returns ALL entries with source\_status='present\_local'
- This includes both annotation\_proxy AND annotation\_source AND owned\_original
- Attaches the local\_path to the eval spec (human\_lovo\_manifest.json)
- Later, extract\_video() calls \_derive\_video\_path(spec)
- If spec has explicit 'video\_path', uses that; else derives from labels path

ISSUE: The eval's labels filenames don't indicate which FILE they were made from:

Example: mahadevpura\_2.rallies.csv -> doesn't say 'proxy' or 'original'

If the trajectory/labels were created from the proxy but extract\_video loads the original (or vice versa), frame indices will be misaligned ? person detection will apply features to the wrong frames in the output candidates JSON.

## 22. assistant (2026-07-04T00:52:14.556Z)

Now let me look at fusion\_golden.py's actual path derivation logic more carefully:

## 23. user (2026-07-04T00:52:22.266Z)

=== fusion\_golden.py path derivation ===

```
def _derive_video_path(spec: Dict[str, Any]) -> str:
 '''Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
 golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming convention).'''
 if spec.get("video_path"):
 return str(spec["video_path"])
 labels = str(spec["labels"])
 if labels.endswith(".rallies.csv"):
 return labels[: -len(".rallies.csv")] + ".MP4"
 raise ValueError(...)
```

When `attach_source_video_paths()` is called:

1. Loads source manifest (PR #62 `golden_source_videos.json`)
2. For each eval spec, looks up `spec.get('name')` in `source_paths`
3. Sets `spec['video_path'] = source_path`

For example, for 'mahadevpura\_2':

- `load_source_video_paths()` returns:  
"mahadevpura\_2": "G:\...\annotation\_proxy.mp4"
- `attach_source_video_paths()` sets:  
`spec['video_path'] = 'G:\...\annotation_proxy.mp4'`
- `extract_video()` calls `_derive_video_path(spec)`
- `_derive_video_path` returns `spec.get('video_path')` ? the proxy

BUT: The fallback is dangerous:

- If `source_manifest` is NOT provided or if lookup fails,  
`spec['video_path']` is never set
- Then `_derive_video_path` falls back to deriving from labels path:  
"C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura\_2.rallies.csv"  
-> "C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura\_2.MP4"
- This derived path probably DOES NOT EXIST (it's in output/, not on G: or F:)
- `extract_video` would throw `FileNotFoundError`

CROSS-REPO BREAK SCENARIO:

1. If #458 runs `fusion_golden` WITHOUT `--source-manifest` parameter
2. Or if `--source-manifest` points to the wrong file
3. Or if the `source_status` was NOT 'present\_local'
4. Then `attach_source_video_paths()` would attach 0 paths
5. `extract_video()` would try to derive from labels path
6. Derived path would be wrong (nonexistent or original vs proxy)
7. `FileNotFoundError` thrown, `fusion_golden` aborts

## 24. assistant (2026-07-04T00:52:24.448Z)

Let me check what the A7 execution record in `ALPHA_LAUNCH_READINESS.md` says about `source_manifest`:

## 25. user (2026-07-04T00:52:24.991Z)

```
179 A7 full-fusion/audio feature refresh record:
180
181 ```powershell
182 python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-
dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
183 python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json
--features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json
--augment
184 ```
185
186 Result: the GPU pass completed without a thermal abort after fans were set to max
performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are
present for `15/15`. The large regenerated clips completed as follows: `mahadevpura_1` `1`
window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`,
`GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill_Doubles` `59` / `12` /
```

```
`43`, `mahadevpura_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton_BXH_2` `29` /
`1` / `44`. The source manifest supplied all 15 source-video paths, including the F-drive
Boxhill originals.
187
188 A8 fusion/default-decision record:
```

## 26. assistant (2026-07-04T00:52:26.989Z)

Perfect. Now I have the full picture. Let me create a final comprehensive analysis:

## 27. user (2026-07-04T00:52:48.368Z)

```
Exit code 2
/usr/bin/bash: -c: line 188: unexpected EOF while looking for matching `''
```

## 28. user (2026-07-04T00:52:57.995Z)

```
[
 {
 "file": "data/golden_source_videos.json",
 "line": "mahadevpura_2, GX030094, Badminton_BXH_2, GX020094",
 "summary": "annotation_proxy entries (5 videos) lack metadata indicating which file eval
labels were made from; frame-index misalignment risk if proxy/original differ in
resolution/fps/codec",
 "failure_scenario": "fusion_golden loads proxy file; eval trajectory/labels created from
original (or vice versa); person detection applies features to misaligned frames; output
candidate indices corrupt"
 },
 {
 "file": "data/golden_source_videos.json",
 "line": "GX010137 and GX010141 notes field",
 "summary": "Notes document lower-res derivatives (GX010137_1080p.mp4, GX010141_1080p.mp4)
with different MD5s; eval labels may have been created from derivatives, not from full-res
originals now being recorded",
 "failure_scenario": "Eval trajectory/labels created from 1080p derivative; fusion_golden
loads full-res original; trajectory parsing fails or produces misaligned shuttlepoints due to
resolution mismatch"
 },
 {
 "file": "backend/eval/fusion_golden.py",
 "line": "135-145",
 "summary": "If attach_source_video_paths fails to set video_path, _derive_video_path falls
back to deriving from labels filename, targeting nonexistent output/ paths",
 "failure_scenario": "Missing/wrong --source-manifest param -> attach attaches 0 paths ->
extract_video tries to load derived nonexistent path -> FileNotFoundError aborts fusion_golden
for all 15"
 },
 {
 "file": "backend/eval/fusion_golden.py",
 "line": "176-180",
 "summary": "File existence check (os.path.isfile) occurs after GPU detector initialization
and trajectory parsing; slow failure if file is wrong or missing",
 "failure_scenario": "Wrong video file path supplied -> detector loads ->
detections_over_windows called on wrong/nonexistent file -> undefined behavior (crash/silent
wrong output/timeout)"
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "186",
 "summary": "A7 claim: 'source manifest supplied all 15 source-video paths'. But manifest
mixes annotation_proxy (5 videos), annotation_source (2), and owned_original (8); no metadata
tags which eval labels come from which role",
 "failure_scenario": "Reviewer cannot verify which video files the eval labels actually
reference; silently loads wrong source file; frame-indexed features corrupt but no error raised"
```

```
}
]
```

### 29. assistant (2026-07-04T00:53:00.118Z)

Perfect. Now let me verify there are no other concrete breaks by checking the actual paths one more time:

### 30. user (2026-07-04T00:53:10.288Z)

```
<stdin>:3: SyntaxWarning: invalid escape sequence '\M'
<stdin>:4: SyntaxWarning: invalid escape sequence '\M'
<stdin>:8: SyntaxWarning: invalid escape sequence '\['
<stdin>:9: SyntaxWarning: invalid escape sequence '\['
<stdin>:10: SyntaxWarning: invalid escape sequence '\['
<stdin>:11: SyntaxWarning: invalid escape sequence '\['
<stdin>:12: SyntaxWarning: invalid escape sequence '\['
<stdin>:13: SyntaxWarning: invalid escape sequence '\['
<stdin>:14: SyntaxWarning: invalid escape sequence '\['
<stdin>:15: SyntaxWarning: invalid escape sequence '\['
Checking for concrete cross-repo breaks:
```

All 15 entries:

- load\_source\_video\_paths will return all 15 (all have source\_status='present\_local')
- attach\_source\_video\_paths will set spec['video\_path'] for all 15 matching names

But we have 3 HIGH-RISK categories for FRAME MISALIGNMENT:

1. annotation\_proxy (5 videos): GX030094, GX020094, Badminton\_BXH\_2, mahadevpura\_2
  - These are proxies (lower quality files for annotation)
  - Eval labels likely made FROM these proxy files
  - Frame indices will match IF fusion\_golden loads same proxy
  - Will BREAK if load priority is wrong (loads original instead)
2. annotation\_source (2 videos): GX010128, mahadevpura\_1
  - Labeled as source role (not proxy)
  - Need to verify eval labels match this file
3. owned\_original with derivative note (2 videos): GX010137, GX010141
  - Notes say: eval run used lower-res derivatives (1080p)
  - But manifest records full-res originals
  - Frame count mismatch CERTAIN if fusions tries full-res

VERDICT: Candidates 1-2 are real, candidates 3-5 are contingent/operational.

### 31. assistant (2026-07-04T00:53:28.340Z)

Now let me finalize the JSON array with only the concrete, actionable breaks:

```
```json  
[  
  {  
    "file": "C:/Users/avidu/Projects/khelsutra-guru/sports-data-  
collector/data/golden_source_videos.json",  
    "line": "GX010137 and GX010141 entries (notes field)",  
    "summary": "Notes document that eval was run against 1080p derivatives (GX010137_1080p.mp4  
MD5: 74eedbc35d739c231344b8929df4b4a7, GX010141_1080p.mp4 MD5:  
daaf8af198640516d14b217f50445d99), but manifest now records full-res originals with different  
MD5s and frame counts",  
    "failure_scenario": "fusion_golden loads full-res original file; TrackNetRunner parses  
trajectory CSV made from 1080p (different frame count); trajectory frame indices misaligned;  
person detection applies features to wrong shuttle positions; output candidates have corrupted  
frame-indexed features"  
  },  
]
```

```

{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/sports-data-
collector/data/golden_source_videos.json",
  "line": "mahadevpura_2, GX030094, Badminton_BXH_2, GX020094 entries",
  "summary": "Five annotation_proxy entries lack explicit link to their counterpart source
file; no field indicates whether eval trajectory/labels were created from proxy or original;
annotation_proxy filenames suggest proxies were used for annotation but this is not formally
recorded",
  "failure_scenario": "If annotation_proxy and original differ in resolution/fps,
fusion_golden loads wrong file; trajectory frame indices misalign; person detection detects on
wrong frame range; output candidates corrupt or feature-null for misaligned window"
},
{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/fusion_golden.py",
  "line": "68-94 (load_source_video_paths function)",
  "summary": "Function loads all entries with source_status='present_local' regardless of
source_role; no filtering or prioritization by source_role (proxy vs original); unclear
precedence if both proxy and original exist for same video",
  "failure_scenario": "If golden_source_videos.json ever adds both proxy and original for same
video_id (e.g., two GX010137 entries), load_source_video_paths returns last one; attaches wrong
file path; fusion_golden loads opposite file from where eval labels came from"
},
{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/ALPHA_LAUNCH_READINESS.md",
  "line": "186",
  "summary": "A7 claim states 'The source manifest supplied all 15 source-video paths,
including the F-drive Boxhill originals', but does not document that 5 videos are proxies and 2
are derivatives; no traceability matrix linking eval labels to specific files in
golden_source_videos.json",
  "failure_scenario": "Reviewer cannot verify that fusion_golden will load the correct source
file for each eval entry; frame-indexed feature corruption remains silent if wrong file is
loaded; no audit trail to detect frame misalignment"
},
{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/fusion_golden.py",
  "line": "135-145 (_derive_video_path function)",
  "summary": "Fallback path derivation from labels filename (X.rallies.csv -> X.MP4) targets
nonexistent files in output/ directory; silent failure mode if attach_source_video_paths does
not set video_path (e.g., due to missing --source-manifest or typo in source_status)",
  "failure_scenario": "PR #458 runs fusion_golden without --source-manifest param;
attach_source_video_paths attaches 0 paths; extract_video calls _derive_video_path which returns
invalid output/X.MP4 path; FileNotFoundError thrown, fusion_golden aborts for all 15 videos"
}
]
...

[
  {
    "file": "C:/Users/avidu/Projects/khelsutra-guru/sports-data-
collector/data/golden_source_videos.json",
    "line": "GX010137 and GX010141 entries (notes field)",
    "summary": "Notes document that eval was run against 1080p derivatives (GX010137_1080p.mp4
MD5: 74eedbc35d739c231344b8929df4b4a7, GX010141_1080p.mp4 MD5:
daaf8af198640516d14b217f50445d99), but manifest now records full-res originals with different
MD5s and frame counts",
    "failure_scenario": "fusion_golden loads full-res original file; TrackNetRunner parses
trajectory CSV made from 1080p (different frame count); trajectory frame indices misaligned;
person detection applies features to wrong shuttle positions; output candidates have corrupted
frame-indexed features"
  },
  {
    "file": "C:/Users/avidu/Projects/khelsutra-guru/sports-data-

```

```

collector/data/golden_source_videos.json",
  "line": "mahadevpura_2, GX030094, Badminton_BXH_2, GX020094 entries",
  "summary": "Five annotation_proxy entries lack explicit link to their counterpart source
file; no field indicates whether eval trajectory/labels were created from proxy or original;
annotation_proxy filenames suggest proxies were used for annotation but this is not formally
recorded",
  "failure_scenario": "If annotation_proxy and original differ in resolution/fps,
fusion_golden loads wrong file; trajectory frame indices misalign; person detection detects on
wrong frame range; output candidates corrupt or feature-null for misaligned window"
},
{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/fusion_golden.py",
  "line": "68-94 (load_source_video_paths function)",
  "summary": "Function loads all entries with source_status='present_local' regardless of
source_role; no filtering or prioritization by source_role (proxy vs original); unclear
precedence if both proxy and original exist for same video",
  "failure_scenario": "If golden_source_videos.json ever adds both proxy and original for same
video_id (e.g., two GX010137 entries), load_source_video_paths returns last one; attaches wrong
file path; fusion_golden loads opposite file from where eval labels came from"
},
{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/docs/ALPHA_LAUNCH_READINESS.md",
  "line": "186",
  "summary": "A7 claim states 'The source manifest supplied all 15 source-video paths,
including the F-drive Boxhill originals', but does not document that 5 videos are proxies and 2
are derivatives; no traceability matrix linking eval labels to specific files in
golden_source_videos.json",
  "failure_scenario": "Reviewer cannot verify that fusion_golden will load the correct source
file for each eval entry; frame-indexed feature corruption remains silent if wrong file is
loaded; no audit trail to detect frame misalignment"
},
{
  "file": "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/fusion_golden.py",
  "line": "135-145 (_derive_video_path function)",
  "summary": "Fallback path derivation from labels filename (X.rallies.csv -> X.MP4) targets
nonexistent files in output/ directory; silent failure mode if attach_source_video_paths does
not set video_path (e.g., due to missing --source-manifest or typo in source_status)",
  "failure_scenario": "PR #458 runs fusion_golden without --source-manifest param;
attach_source_video_paths attaches 0 paths; extract_video calls _derive_video_path which returns
invalid output/X.MP4 path; FileNotFoundError thrown, fusion_golden aborts for all 15 videos"
}
]

```